

SESA3029 - Method of Characteristics Coursework

Veer Phatak – 35084553

$$M_{des} = 2.3$$

Summary:

This project aims to design a minimum length supersonic nozzle using the method of characteristics (MoC), a computational method used to estimate the nozzle size. First, a Prandtl-Meyer function and Mach angle are computed for $M_{des}=2.3$, which take this Mach number and return the values seen later. Then the MoC is used, where different angles and Mach numbers are calculated for 5 'characteristic lines', which allow us to trace the shape of the nozzle and observe the angles and flow speed throughout. These values then allow us to calculate the x and y co-ordinates along these lines and hence the minimum length nozzle shape is obtained.

Method of Characteristics:

The goal of this method for nozzle design is to create a smooth area expansion that uniformly accelerates flow from an initial Mach number - based on the initial θ - to M_{des} , without shocks. The design starts at the expansion corner of the nozzle, where the flow is linearly increasing outwards to expand smoothly. 5 characteristic lines have been chosen to each model an expansion wave and these lines reflect between the nozzle centreline and the wall. By calculating properties at each intersection, the flow properties of each characteristic line and the nozzle wall co-ordinates can be obtained. MoC does this through characteristic lines, which quantify fixed points throughout the flow field called Riemann invariants, as seen below.

$$R_A^+ = v_A - \theta_A \quad (1)$$

$$R_B^- = v_B + \theta_B \quad (2)$$

The θ is the local flow angle and the v is the Prandtl-Meyer function which is calculated using the following equation:

$$v(M) = \int \frac{\sqrt{M^2 - 1}}{1 + \frac{\gamma - 1}{2} M^2} \frac{dM}{M} \quad (3)$$

As the characteristic lines interact throughout the nozzle, each intersection can be treated as the focal point 'P', making points A and B the closest reference points below and above 'P' respectively, allowing v and θ for point P to be calculated from A and B. The characteristic lines exit each point at a Mach angle of μ relative to the flow. The Mach number is calculated based on interpolation from the IFT I made for $\gamma = 1.667$, for helium.

$$\mu = \arcsin\left(\frac{1}{M}\right) \quad (4)$$

When the θ , μ , and v have been calculated for the points where the characteristics lines interact with the wall, centreline, or each other, the x and y co-ordinates can be calculated using the following set of equations:

$$\alpha_{AP} = \frac{1}{2} [(\theta + \mu)_A + (\theta + \mu)_P] \quad (5)$$

$$\alpha_{BP} = \frac{1}{2} [(\theta - \mu)_B + (\theta - \mu)_P] \quad (6)$$

$$\frac{y_P - y_A}{x_P - x_A} = \tan \alpha_{AP} \quad (7)$$

$$\frac{y_P - y_B}{x_P - x_B} = \tan \alpha_{BP} \quad (8)$$

$$x_P = \frac{x_B \tan \alpha_{BP} - x_A \tan \alpha_{AP} - (y_B - y_A)}{\tan \alpha_{BP} - \tan \alpha_{AP}} \quad (9)$$

$$y_P = y_A + (x_P - x_A) \tan \alpha_{AP} \quad (10)$$

Equations 7 and 8 shows the gradient of the characteristic lines, which is calculated by finding the tan of the average of the angles between 2 points, AP and BP respectively. By rearranging equations 7 and 8, equation 9 is obtained and this solves for the x co-ordinate of the intersection point. Equation 7 is rearranged into equation 10 to solve for the y co-ordinate of the intersection point.

These equations apply to general interactions, but conditions are slightly different at the centreline or wall points. For the centreline, only one side of the symmetry plane needs to be calculated – α_{BP} , as centreline acts as a mirror so what happens for the upper side happens for the lower side, so only half the calculations need to be made. This results in the following equations needing to be used instead:

$$y_P = y_0 \quad (11)$$

$$x_P = x_B - \frac{y_B - y_0}{\tan \alpha_{BP}} \quad (12)$$

For the wall points, point B becomes the previous point along the wall as the nozzle is a simple region of flow, so the characteristic lines are not reflected off the wall points. This is due to flow angle being equal to the wall angle, meaning there is not turn in flow at the wall, so the equations used are:

$$\alpha_{AP} = (\theta + \mu)_A \quad (13)$$

$$\alpha_{BP} = \frac{1}{2} (\theta_B + \theta_P) \quad (14)$$

Mach Angle & Prandtl-Meyer Computation and Verification:

$$\mu = 25.771462^\circ$$

$$v = 27.769673^\circ$$

This was computed in python using the code seen in figure 1 below. The code was verified by changing gamma to 1.4(for air) and obtaining a v of 34.28279°, which corresponds to the value of 34.2828° given in the IFT. Additionally, changing the Mach number returns a μ of 90° and a v of 0°, which also corresponds to the IFT.

```

import numpy as np

def prandtl_meyer(M, gamma):
    # degrees
    nu = np.sqrt((gamma + 1)/(gamma - 1)) * np.arctan(np.sqrt((gamma - 1) * (M**2 - 1) / (gamma + 1)))
    nu -= np.arctan(np.sqrt(M**2 - 1))
    return np.degrees(nu)

def mach_angle(M):
    return np.degrees(np.arcsin(1/M))

M_exit = 2.3
gamma = 1.667
mu = mach_angle(M_exit)
nu_exit = prandtl_meyer(M_exit, gamma)
print(f"Mach angle (mu): {mu: 4f}degrees")
print(f"Prandtl-Meyer function (nu_exit):{nu_exit:4f}degrees")

```

Figure 1: Prandtl-Meyer Function and Mach Angle Code

Method of Characteristics Template & Calculations:

Poin	R+	R-	θ(deg)	v(deg)	M	μ(deg)	θ+μ	θ-μ	α(AP)	α(BP)	Xa	Ya	Xb	Yb	x	y
0a	N/A	0.3000	0.1500	0.1500	1.0237	77.6521	77.8021	-77.5021							0.0000	1.0000
0b	N/A	7.1674	3.5837	3.5837	1.2205	55.0200	58.6037	-51.4363							0.0000	1.0000
0c	N/A	14.0348	7.0174	7.0174	1.3683	46.9572	53.9746	-39.9398							0.0000	1.0000
0d	N/A	20.9023	10.4511	10.4511	1.5095	41.4899	51.9410	-31.0387							0.0000	1.0000
0e	N/A	27.7697	13.8848	13.8848	1.6515	37.2663	51.1512	-23.3815							0.0000	1.0000
1	0.3000	0.3000	0.0000	0.3000	1.0386	74.3394	74.3394	-74.3394	N/A	-1.3251	N/A	N/A	0.0000	1.0000	0.2508	0.0000
2	0.3000	7.1674	3.4337	3.7337	1.2272	54.5711	58.0048	-51.1374	1.1549	-0.8951	0.2508	0.0000	0.0000	1.0000	0.4464	0.4430
3	0.3000	14.0348	6.8674	7.1674	1.3745	46.6799	53.5473	-39.8125	0.9735	-0.6960	0.4464	0.4430	0.0000	1.0000	0.5263	0.5603
4	0.3000	20.9023	10.3011	10.6011	1.5156	41.2845	51.5857	-30.9834	0.9175	-0.5412	0.5263	0.5603	0.0000	1.0000	0.5909	0.6448
5	0.3000	27.7697	13.7348	14.0348	1.6578	37.1014	50.8362	-23.3665	0.8938	-0.4080	0.5909	0.6448	0.0000	1.0000	0.6504	0.7189
6	0.3000	N/A	13.7348	14.0348	1.6578	37.1014	50.8362	-23.3665	0.8873	0.2410	0.6504	0.7189	0.0000	1.0000	1.0996	1.2703
7	7.1674	7.1674	0.0000	7.1674	1.3745	46.6799	46.6799	-46.6799	N/A	-0.8536	N/A	N/A	0.4464	0.4430	0.8328	0.0000
8	7.1674	14.0348	3.4337	10.6011	1.5156	41.2845	44.7182	-37.8508	0.8575	-0.6777	0.8328	0.0000	0.5263	0.5603	0.9927	0.1849
9	7.1674	20.9023	6.8674	14.0348	1.6578	37.1014	43.9688	-30.2339	0.8339	-0.5342	0.9927	0.1849	0.5909	0.6448	1.1239	0.3294
10	7.1674	27.7697	10.3011	17.4685	1.8046	33.6508	43.9519	-23.3496	0.8272	-0.4077	1.1239	0.3294	0.6504	0.7189	1.2457	0.4618
11	7.1674	N/A	10.3011	17.4685	1.8046	33.6508	43.9519	-23.3496	0.7671	0.2098	1.2457	0.4618	1.0996	1.2703	2.3634	1.5393
12	14.0348	14.0348	0.0000	14.0348	1.6578	37.1014	37.1014	-37.1014	N/A	-0.6541	N/A	N/A	0.9927	0.1849	1.2339	0.0000
13	14.0348	20.9023	3.4337	17.4685	1.8046	33.6508	37.0845	-30.2171	0.7073	-0.5275	1.2339	0.0000	1.1239	0.3294	1.4185	0.1578
14	14.0348	27.7697	6.8674	20.9023	1.9590	30.6955	37.5629	-23.8281	0.7114	-0.4117	1.4185	0.1578	1.2457	0.4618	1.5945	0.3095
15	14.0348	N/A	6.8674	20.9023	1.9590	30.6955	37.5629	-23.8281	0.6556	0.1498	1.5945	0.3095	2.3634	1.5393	3.3963	1.6953
16	20.9023	20.9023	0.0000	20.9023	1.9590	30.6955	30.6955	-30.6955	N/A	-0.5316	N/A	N/A	1.4185	0.1578	1.6869	0.0000
17	20.9023	27.7697	3.4337	24.3360	2.1232	28.0979	31.5316	-24.6642	0.6030	-0.4232	1.6869	0.0000	1.5945	0.3095	1.9221	0.1619
18	20.9023	N/A	3.4337	24.3360	2.1232	28.0979	31.5316	-24.6642	0.5503	0.0899	1.9221	0.1619	3.3963	1.6953	4.5977	1.8035
19	27.7697	27.7697	0.0000	27.7697	2.3000	25.7715	25.7715	-25.7715	N/A	-0.4401	N/A	N/A	1.9221	0.1619	2.2660	0.0000
20	27.7697	N/A	0.0000	27.7697	2.3000	25.7715	25.7715	-25.7715	0.4498	0.0300	2.2660	0.0000	4.5977	1.8035	6.0945	1.8484

Table 1: MoC computed values

Point 1 Calculations:

$$R_1^+ = R_1^- = R_{0a}^- = 0.3^\circ$$

$$\theta_1 = 0^\circ$$

$$v_1 = R_1^+ + \theta_1 = 0.3^\circ$$

Using interpolation from the IFT for helium: $M_1 = 1.0386$

Using equation 4:

$$\mu_1 = \frac{1}{M} = 74.3394^\circ$$

Using equation 6:

$$\alpha_{0a,1} = -1.3251 \text{ rad}$$

Using equation 11:

$$y_P = y_0 = 0$$

Using equation 12:
$$x_1 = 0 - \frac{1-0}{\tan(-1.3251)} = 0.2508$$

Therefore, the co-ordinates for point 1 are (0.2508,0) showing that it is on the centreline.

Point 2 Calculations:

$$R_2^+ = R_1^+ = 0.3^\circ$$

$$R_2^- = R_{0b}^- = 7.1674^\circ$$

$$\theta_2 = \frac{R_2^- - R_2^+}{2} = 3.4337$$

$$\nu_2 = \frac{R_2^+ + R_2^-}{2} = 3.7337^\circ$$

Using interpolation from the IFT for helium: $M_2 = 1.2272$

$$\mu_2 = \frac{1}{M_2} = 54.5711^\circ$$

$$\alpha_{12} = \frac{1}{2}(74.3394 + 58.0048) = 1.1549 \text{ rad}$$

$$\alpha_{0b,2} = \frac{1}{2}(-51.4363 + -51.1374) = -0.8951 \text{ rad}$$

Using equation 9: $x_2 = 0.4464$

Using equation 10: $y_2 = 0.4430$

Therefore, the co-ordinates for point 2 are (0.4464,0.4430)

Point 20 Calculations:

$$R_{20}^+ = R_{19}^+ = 27.7697^\circ$$

$$\theta_{20} = \theta_{19} = 0^\circ$$

$$\nu_{20} = R_{20}^+ + \theta_{20} = 27.7697^\circ$$

Using interpolation from the IFT for helium: $M_{20} = 2.3 = M_{des}$ as point 20 is the exit point

$$\mu_{20} = \frac{1}{M_{20}} = 25.7715^\circ$$

Using equation 13: $\alpha_{19,20} = (\theta + \mu)_{19} = 0.4498 \text{ rad}$

Using equation 14: $\alpha_{18,20} = \frac{1}{2}(\theta_{18} + \theta_{20}) = 0.03 \text{ rad}$

Using equation 9: $x_2 = 6.0944$

Using equation 10: $y_2 = 1.8484$

Therefore, the co-ordinates for point 20 are (6.0944, 1.8484)

Results:

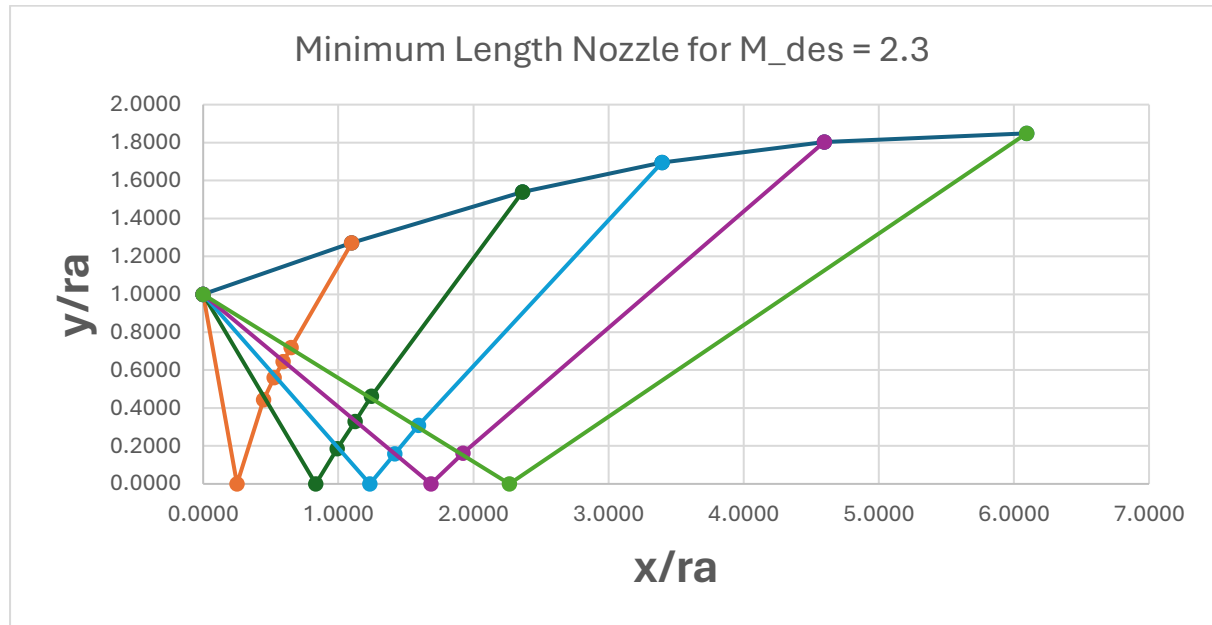


Fig 2. Plot of the MoC nozzle design with nozzle wall and characteristic lines highlighted

The nozzle shape and flow lines seen in Figure 2 are consistent with the data from Table 1, and the flow angle at points 19 and 20 being 0° shows how the flow does not accelerate at these points, hence the flow field has reached the designated Mach number of 2.3 at the exit point 20. The characteristic lines not reflecting at the wall lines show how the flow behaved as expected in a minimum length supersonic nozzle.

For Quasi-1D theory, the area ratio can be calculated using the following equation:

$$\frac{A}{A^*} = \frac{1}{M_{des}} \times \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M^2 \right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}} = 1.8672$$

The obtained A/A^* from the method of characteristics is 1.8484, so the percentage error between MoC and quasi 1D theory is $\frac{1.8484 - 1.8672}{1.8672} \times 100 = -1.007\%$, which is a very small percentage error and so the method of characteristics can be used to very accurately predict minimum length nozzle area ratios. The method of characteristics gets more accurate as the more characteristic lines are used, the closer the y/ra at 20 tends to A/A^* , so a more refined computation provides a more accurate nozzle design prediction.

The main limitation of method of characteristics is the refinement of the characteristic lines, as using more characteristic allows for a more detailed prediction of the flow field of the nozzle, and this allows for more accurate positions for the nozzle wall and intersection points to be mapped. The other prominent limitation of the method is calculating the Mach number based on the Prandtl-Meyer function and then linearly interpolating from the generated IFT for $\gamma = 1.667$, which means that it is highly likely that there are small errors for each Mach number, but it is an almost negligible error. Overall, these limitations do not hinder the accuracy of method of characteristics to a sufficient degree that it invalidates the method, as method of characteristics, when compared to quasi-1D theory, is an accurate method for prediction.